

Machine Learning Based Modeling for Diabetes Prediction

Chandra Sekhar Dammalapati
Department of Computer Science
SUNY Oswego
Oswego, NY, United States
cdammala@oswego.edu

Xiaoliang Wang
Department of Computer Science
SUNY Oswego
Oswego, NY, United States
xiaoliang.wang@oswego.edu

Abstract—Diabetes Mellitus (DM) is among the most widespread chronic diseases worldwide and can cause serious health complications if not identified at an early stage. The timely diagnosis and proper intervention play a crucial role in reducing risks such as cardiovascular disorders and kidney damage. The dataset includes both demographic and clinical variables such as age, gender, Body Mass Index (BMI), HbA1c levels, blood glucose levels, and existing health conditions. Several algorithms, including Logistic Regression, Random Forest, Support Vector Machine (SVM), Decision Tree, and Gradient Boosting, are implemented and evaluated using metrics, such as accuracy, precision, recall, F1-score, and ROC-AUC. The findings indicate that ensemble models outperform other methods, demonstrating the effectiveness of data-driven approaches in enhancing early diabetes detection and supporting healthcare decision-making.

Index Terms—Diabetes, Machine Learning, Classification

I. INTRODUCTION

Elevated blood glucose levels are a hallmark of Diabetes Mellitus (DM), a chronic metabolic disease caused by either inadequate insulin synthesis or poor insulin utilization. It now affects millions of people of all ages and has grown to be a significant global public health concern. Diabetes is becoming more common, which has increased the risk of associated health issues, such as kidney failure, cardiovascular disease, and nerve damage. This highlights the critical need for early detection and preventative healthcare measures. The potential of data-driven technology to improve early diagnosis and intervention is highlighted by the fact that traditional diagnostic approaches, although successful, can be time-consuming and dependent on clinical experience. Recent developments in machine learning and data science have created new opportunities for predictive healthcare analytics [1]–[4]. Machine learning algorithms are able to properly categorize people as diabetic or non-diabetic by utilizing patient data, which includes clinical, demographic, and lifestyle aspects of information. The purpose of this research is to use machine learning techniques to create a reliable predictive model for the classification of diabetes. The 100,000 individual records in the data set utilized in this study include a variety of characteristics, including age, gender, BMI, HbA1c level, blood glucose level, smoking history, and comorbidities like heart disease and hypertension. Through systematic data preprocessing, the application of multiple classification algorithms and hyperparameter tuning,

this work seeks to identify the optimal hyperparameter settings and evaluate the performance of various predictive models. The goal is to contribute to the development of intelligent healthcare solutions that support clinicians in early detection of diabetes and effective patient management.

II. METHODOLOGY

A. DATASET

- **Dataset:** The dataset was obtained from publicly available platform Kaggle [5]. The dataset includes essential demographic variables such as age, gender, race, and location, which help analyze how diabetes prevalence varies across different population groups. Clinical attributes such as Body Mass Index (BMI), HbA1c, and blood glucose levels provide quantitative insights into metabolic health. Additionally, information regarding hypertension, heart disease, and smoking history captures comorbidities and lifestyle factors that are commonly linked with diabetes onset. These attributes were carefully chosen to ensure that the dataset encompasses both direct and indirect predictors of diabetes.

B. DATA PREPROCESSING

- **Handling Duplicate Records:** Duplicate entries in the dataset can lead to biased model performance and incorrect analysis. Therefore, the dataset is checked for duplicate rows using *Pandas* function *duplicated()*. If duplicates are found, they can be removed to ensure that each record represents a unique individual. This step improves data integrity and prevents the model from learning redundant patterns.
- **Encoding Categorical Variables:** The categorical variable *SmokingHistory* was transformed using one-hot encoding, creating binary indicator variables for each category: *Former*, *Never*, *Current*, *Not Current* and *Ever*.
- **Feature Selection:** Variables such as *Year* and *Location*, which represent when and where the data were collected rather than patient-specific clinical characteristics, were removed from the feature set. Excluding these variables helped reduce unnecessary complexity and

minimize the risk of the model learning dataset-specific patterns that may not generalize to new data.

- **Handling Class Imbalance:** Since the dataset contained more non-diabetic cases than diabetic cases, Synthetic Minority Over-sampling Technique (SMOTE) was applied on training set only after each train-test split, to balance the training set by generating synthetic samples for the minority class, improving the model's prediction accuracy.

C. MODEL DEVELOPMENT AND EVALUATION

- **Hyperparameter Tuning:** Hyperparameter optimization was conducted using *GridSearchCV* function from *Scikit – learn* library with 5-fold cross-validation. The best hyperparameter settings were selected based on ROC-AUC performance. ROC-AUC was selected as the primary metric because it evaluates a model's ability to distinguish between diabetic and non-diabetic patients across all classification thresholds and is less sensitive to class imbalance than accuracy alone. As shown in Table I, optimal hyperparameter settings are highlighted with bold.

TABLE I
HYPERPARAMETER TUNING

Model	Hyperparameter
Random Forest	classifier__n_estimators: [100, 300, 500] classifier__max_depth: [5, 10, None] classifier__min_samples_split: [2, 5] classifier__min_samples_leaf: [1, 2]
Gradient Boosting	classifier__n_estimators: [100, 300, 500] classifier__learning_rate: [0.01, 0.05, 0.1] classifier__max_depth: [3, 5, 7]
SVM	classifier__C: [0.1, 1, 10] classifier__gamma: [<i>'scale'</i> , 0.01]
Decision Tree	classifier__max_depth: [3, 5, 10 , <i>None</i>] classifier__min_samples_split: [2, 5] classifier__min_samples_leaf: [1, 2]
Logistic Regression	classifier__C: [0.01 , 0.1, 1, 10] classifier__solver: [<i>'liblinear'</i> , <i>'lbfgs'</i>]

- **Model Development:** Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and Gradient Boosting models were developed based on the optimal hyperparameter settings. Each model was trained using the preprocessed training dataset and optimized using 5-fold cross-validation. Ensemble methods were emphasized for their strong performance in handling complex, non-linear relationships in the data.
- **Model Evaluation:** The performance of each machine learning model was evaluated using metrics such as accuracy, precision, recall, F1 Score, and ROC AUC. The models were compared to identify the most accurate and reliable algorithm for diabetes prediction.

III. EXPERIMENT RESULTS

As shown in Table II, Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and Gradient Boosting were trained and evaluated using metrics including accuracy, precision, recall, F1-score, and ROC-AUC. To

improve reliability, 5-fold cross-validation was also applied, ensuring that the results were consistent and not dependent on a single train-test split. Gradient Boosting produced the best overall performance in this study. The model achieved the highest ROC-AUC score of 0.9781, along with a accuracy of 0.9718, precision of 0.9703, recall of 0.6893 and F1-score of 0.8059. Gradient Boosting achieved the highest ROC-AUC and precision among all evaluated models, although recall indicates that some diabetic cases remain difficult to identify. Gradient Boosting provided a better balance between precision and overall performance. The results highlight the importance of machine learning in healthcare for early disease detection, timely intervention, and improved patient outcomes.

TABLE II
EXPERIMENTAL RESULTS OF PREDICTION MODELS

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Random Forest	0.9678	0.9024	0.6968	0.7864	0.9714
Gradient Boosting	0.9718	0.9703	0.6893	0.8059	0.9781
SVM	0.8862	0.4206	0.8945	0.5721	0.9624
Decision Tree	0.9160	0.5069	0.8226	0.6262	0.9646
Logistic Regression	0.8870	0.4213	0.8809	0.5700	0.9621

IV. CONCLUSION

This study developed and evaluated several machine learning models for diabetes prediction using demographic and clinical health data. Among the evaluated models, Gradient Boosting achieved the best overall performance, demonstrating its effectiveness in distinguishing diabetic and non-diabetic individuals. The results highlight the potential of machine learning techniques to support early diabetes detection and assist healthcare professionals in making informed decisions. A limitation of this study is that the models were evaluated using a single publicly available dataset. Future work will focus on validating the proposed approach using independent clinical datasets and exploring additional features and advanced machine learning techniques to improve predictive performance and generalizability.

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